

JEE MAIN REVISION MASTER: STATISTICS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Mean, Variance, Standard Deviation & Coefficient of Variation	10 PYQs	Statistics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mean, Variance, Standard Deviation & Coefficient of Variation	$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2, \quad \text{New Variance after } k = k^2 \sigma^2, \quad \text{C.V.} = \frac{\sigma}{\bar{x}} \times 100$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mean, Variance, Standard Deviation & Coefficient of Variation (High Yield — 10 PYQs)

Core Principle: Variance $\sigma^2 = (\sum x_i^2 / n) - (\bar{x})^2$. Standard deviation $\sigma = \sqrt{\text{Variance}}$. Coefficient of variation C.V. = $\frac{\sigma}{\bar{x}} \times 100$ (measures relative variability; series with smaller C.V. is more consistent). Shift of origin: Adding/subtracting a constant k to each observation changes mean by $\pm k$, but leaves variance and standard deviation completely UNCHANGED. Change of scale: Multiplying each observation by k multiplies mean by k , standard deviation by $|k|$, and variance by k^2 .

■ **Caution / JEE Pitfall:** Adding or subtracting a number to all observations has ZERO effect on standard deviation and variance!

■ **Speed Trick:** Variance of first n natural numbers: $\sigma^2 = (n^2 - 1) / 12$. A classic 5-second question in KCET.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The mean and standard deviation of 100 items are 50 and 4, respectively then the sum of all squares of the items is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The mean deviation about the mean for the data 4,7,8,9,10,12,13,17 is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Let a, b, c, d and e be the observations with mean m and standard deviation S . The standard deviation of the observations $a+k, b+k, \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The mean and standard deviation of 100 items are 50 and 4 , respectively then the sum of all squares of the items is

- (A) 250000
- (B) 251600
- (C) 256100
- (D) 265100

Correct Answer: None

Solution Analysis: For 100 items, the mean is $\bar{x}=50$ and the standard deviation is $\sigma=4$. Using the formula $\sigma^2=\frac{\sum x_{(i)}^2-n\bar{x}^2}{n}$ Substitute the given values: $4^2=\frac{\sum x_{(i)}^2-100 \times 50^2}{100}$ $16=\frac{\sum x_{(i)}^2-2500}{100}$ So, $\frac{\sum x_{(i)}^2-2500}{100}=16$ $\sum x_{(i)}^2=2516 \times 100=251600$ Hence, the sum of all squares of the items is 251600 Correct option: Option B.